

High-precision beam position and size measurement method based on improved Gaussian fitting

Dong Li, *Ping Tan, Yinjie Lin

Abstract -- In pencil beam scanning (PBS) proton therapy, accurate monitoring of the center position and transverse size of the proton beam at the exit of the scanning nozzle is essential to ensure the safety of treatment. Since the beam spots are very small relative to the detected area in clinical applications, an improved Gaussian fitting method was proposed in this study, which extracts only the meaningful strip data beyond the specified noise threshold according to the average power of the background noise. This not only improves the accuracy, but also reduces the amount of computation. By simulation test in MATLAB, when the signal-to-noise ratio (SNR) is higher than 20dB, the accuracy of the beam position is better than 0.2mm, and the relative error of the beam size is better than 2%, and the computing time of algorithm is less than 0.4ms. Through experiments, the accuracy of the proposed algorithm in fitting the beam position was verified, and it is demonstrated that this method still works when the beam irradiates at the edge of ionization chamber.

Index Terms-- Beam monitoring, Gaussian fitting, Proton therapy

I. INTRODUCTION

PROTON therapy is one of the most effective radiation therapy methods for cancers [1]. Huazhong University of Science and Technology is developing a proton therapy facility (HUST-PTF) based on superconducting cyclotron, in which the scanning nozzle locates at the end of beam line and closes to the patient [2]. The scanning nozzle will deliver the proton beam to the target area. So, monitoring the position and size of beam at the exit of nozzle is crucial to maximize the therapy efficacy and to reduce possible toxicities [3].

Ionization chambers (ICs), diode arrays, and scintillating and optical fiber detectors are usually used to monitor the beam [4]-[5]. Compared with other detectors, ionization chamber has the advantage of almost non-destructive to proton beam. Two strip ionization chambers are used in the scanning nozzle in HUST-PTF, which are placed vertically to the beam direction. The strip ionization chamber is mainly composed of an anode plate and two cathode plates which are segmented into 128 strips with a width of 2mm along two

directions perpendicular to each other. The beam irradiation direction is set as Z-direction, and the two directions of the ionization chamber plane are X-direction and Y-direction respectively, as shown in Fig. 1. When the proton beam passes through the ionization chamber, the gas will be ionized, and the generated positive ions and electrons will move to the cathode and anode respectively under the action of the electric field and be collected [6]. The peripheral readout electronics have integrating amplifiers for each strip channel which accumulates the incoming current as charge on a small capacitor for a defined integration period. The integral charge shows a Gaussian distribution in the beam irradiation region, and the center position and size of the beam can be fitted by processing the charge data.

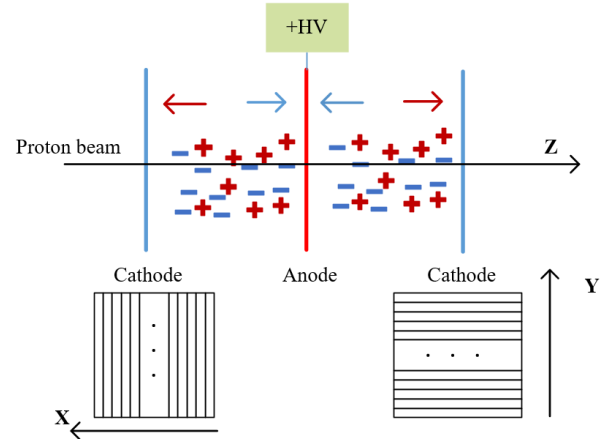


Fig. 1. Schematic diagram of strip ionization chamber.

Typically, the methods for beam position verification are centroid method and Gaussian fitting method. The centroid method makes use of the symmetry of gaussian function, so it can determine the peak position of Gaussian very effectively [7]. But it cannot estimate the width of a peak. Caruana proposed a simple Gaussian fitting algorithm by taking advantage of the property that gaussian function is the exponential of quadratic function [8]. However, this method is very sensitive to noise signal, which result in a large error. Hongwei Guo improved the method of Caruana by applying weighted least square method and iterative method. [9]. But this method needs several iterations to improve accuracy, which make the calculation very complicated. Due to the small size of the beam spot relative to the sensitive area of the strip ionization chamber, only the signals in the beam irradiation area are meaningful, and the rest are noise signals. To extract these meaningful signals, threshold is set based on the average power of the noise. In this way, in the case of $\sigma=10\text{mm}$, the amount of fitting data can be reduced from 128 to less than 30 after processing, which reduces the amount of calculation. When the SNR is 20dB, if set noise threshold,

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the root mean square error of the beam position is less than 0.2mm, which is much more accurate than 6.36mm without the threshold.

II. ALGORITHM AND SIMULATION

This section mainly introduces the improvement process of the traditional Gaussian method and the simulation test of the improved algorithm.

A. Improved Gaussian Fitting Algorithm

Taking advantage of the fact that gaussian function is the exponential of quadratic function, taking the logarithm of (1) can get (2) [7].

$$y = Ae^{-(x-\mu)^2/2\sigma^2} \quad (1)$$

$$\begin{aligned} \ln y &= \ln A - (x - \mu)^2 / 2\sigma^2 \\ &= (\ln A - \mu^2 / 2\sigma^2) + (\mu / \sigma^2)x + (-1 / 2\sigma^2)x^2 \quad (2) \\ &= a + bx + cx^2 \end{aligned}$$

where $a = \ln A - \mu^2 / 2\sigma^2$, $b = \mu / \sigma^2$, and $c = -1 / 2\sigma^2$. Then the nonlinear equation with the unknowns A , μ and σ is transformed into a linear equation with the unknowns a , b and c .

Due to the influence of random noise, the observed value can be modeled as in (3),

$$\hat{y} = y + \eta \quad (3)$$

where η is random noise, y is the true value and $\hat{y}$ is the observed value. Accordingly, the error function is as in (4).

$$\delta = \ln(\hat{y}) - (a + bx + cx^2) \quad (4)$$

Expanding (4) into Taylor series and reasonably omitting the high-order terms,

$$\delta \approx \ln(y) - (a + bx + cx^2) + \eta / y \quad (5)$$

Since the expectation of random noise is zero, the expectation of δ^2 is (6).

$$E(\delta^2) = [\ln(y) - (a + bx + cx^2)]^2 + (\sigma_\eta^2 / y^2) \quad (6)$$

where σ_η is the standard deviation of the noise.

From the second term of (6), it is obvious that when the observation amplitude y is small, the noise will cause a large error in the estimates. In order to eliminate the influence of the amplitude of the observed data, we use weighted least squares to perform Gaussian curve fitting. Remodel the error equation, then expand it into Taylor series and reasonably omitting the high-order terms.

$$\begin{aligned} \delta' &= y\delta = y[\ln(\hat{y}) - (a + bx + cx^2)] \\ &= y[\ln(y + \eta) - (a + bx + cx^2)] \quad (7) \\ &\approx y[\ln(y) - (a + bx + cx^2)] + \eta \end{aligned}$$

so, the expectation of δ'^2 is (8).

$$E(\delta'^2) = y^2[\ln(y) - (a + bx + cx^2)]^2 + \sigma_\eta^2 \quad (8)$$

In (8), the influence of y on the second term is eliminated. And the sum of δ'^2 is (9)

$$\sum \delta'^2 = \sum y_i^2 [\ln(y_i) - (a + bx + cx^2)]^2 \quad (9)$$

where the weight is y_i^2 .

By minimizing the sum of squares of errors, the least square best estimation can be achieved. Differentiating the sum of δ'^2 with respect to a , b , and c and setting the resulting expression to zero, a system of linear equations is obtained, as in (10) and (11).

$$\begin{aligned} \partial(\delta'^2) / \partial a &= 2y^2[\ln(y) - (a + bx + cx^2)] = 0 \\ \partial(\delta'^2) / \partial b &= 2xy^2[\ln(y) - (a + bx + cx^2)] = 0 \quad (10) \\ \partial(\delta'^2) / \partial c &= 2x^2y^2[\ln(y) - (a + bx + cx^2)] = 0 \end{aligned}$$

$$\begin{bmatrix} \sum \hat{y}^2 & \sum x\hat{y}^2 & \sum x^2\hat{y}^2 \\ \sum x\hat{y}^2 & \sum x^2\hat{y}^2 & \sum x^3\hat{y}^2 \\ \sum x^2\hat{y}^2 & \sum x^3\hat{y}^2 & \sum x^4\hat{y}^2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \sum \hat{y}^2 \ln(\hat{y}) \\ \sum \hat{y}^2 x \ln(\hat{y}) \\ \sum \hat{y}^2 x^2 \ln(\hat{y}) \end{bmatrix} \quad (11)$$

Since the true value y is unknown, we have to use $\hat{y}$ instead of y as the weight. By solving (11), the values of parameters a , b and c can be obtained, and then Gaussian parameters A , μ and σ can be calculated.

Considering the small size of the beam spot relative to the detection area of the strip ionization chamber in clinical application, directly using the above algorithm to fit not only increases the amount of calculation, but also has a large error due to the influence of noise signal. In order to obtain more accurate fitting results, we set a threshold value according to the noise signal power to extract meaningful signals, which also reduces the amount of calculation. And this method still works when the beam irradiates at the edge of ionization chamber.

B. SNR Discussion

In a typical proton therapy, the beam current intensity is of the order of nA, and the background noise current of a single channel is 200-300 fA. The SNR of the signal without integration and without considering the gain of the ionization chamber is discussed as (12)

$$SNR_i = 20 \lg(I_s / (N * I_{noise})) \quad (12)$$

where N is the number of effective read channels after data preprocessing, I_s is the intensity of beam current, and I_{noise} is the average noise current of each channel.

Assume that $I_{noise} = 250$ fA and $N = 100$. If $I_s = 0.25$ nA (which is below the lower limit of the beam in HUST_PTF), then $SNR_i = 20$ dB. Due to the effect of integral amplification

and ionization chamber gain, the final SNR will be higher than this value.

C. Simulation

In order to screen out meaningful signals, it is necessary to obtain the average power (P_{noise}) of noise in a period of time according to (13).

$$P_{\text{noise}} = \sum_{i=1}^N (Y_i)^2 / N \quad (13)$$

where Y is the noise signal. The root mean square error of noise can be calculated according to the average power of noise signal. The noise amplitude higher than $3\sigma_{\text{noise}}$ is an extremely unlikely event. Then set the signal extraction threshold w as (14),

$$w = 3\sigma_{\text{noise}} \approx 3\sqrt{P_{\text{noise}}} \quad (14)$$

In order to remove individual noise signals over w and possibly damaged strip signals, a signal is considered to be meaningful only when three consecutive signals near the signal are higher than w .

Using the above threshold and changing the beam size, the relationship between fitting parameter error and SNR is shown in Fig. 2 and Fig. 3. It can be seen that for different beam sizes, when the SNR is higher than 20dB, the accuracy of the beam position is better than 0.2mm, and the relative error of the beam size is better than 2%.

The dashed line in the figure is the result after 8 iterations of Guo's algorithm[9]. Guo improves the fitting accuracy through continuous iteration by fitting all the data and resetting the weights according to the previous fitting results, which has an advantage in large beam spots. However, with the increase of the number of iterations, the computation of this algorithm is very large. And the new algorithm has obvious advantages in the case of low SNR.

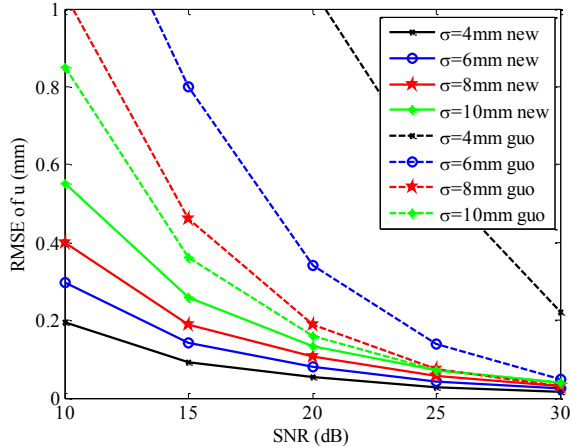


Fig. 2. The relationship between root mean square error of μ , SNR and σ

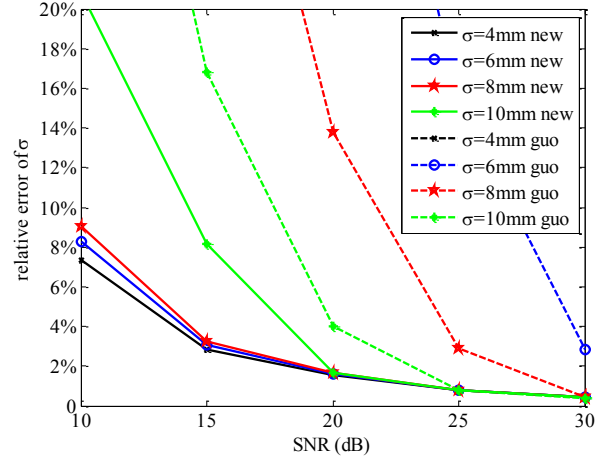


Fig. 3. The relationship between relative error of σ , SNR and σ

III. EXPERIMENT

In order to verify the accuracy of the beam position fitting results of the algorithm, we conducted two experiments in this section. In the first one, the beam spot is located in the sensitive area of the ionization chamber to verify the accuracy of the algorithm under normal conditions. In the second one, the beam spot is located at the edge of the ionization chamber to verify the robustness of the algorithm.

A. Experimental Setup

The experimental setup consists of an X-ray source, a strip ionization chamber, two readout electronics, a medical X-ray film and a PC, as shown in Fig. 4 and Fig. 5. The X-ray source replaces the proton beam, and the medical X-ray film can record the shape of the ordinary stream, which is used to assist in verifying the central position of the beam spot. PC is used to collect experimental data.

The X-ray source is placed perpendicular to the strip ionization chamber. The medical X-ray film is placed close to the lower right edge of the ionization chamber. When the X-ray source is on, the X-rays pass vertically through the medical X-ray film and strip ionization chamber. The peripheral readout electronics integrate the current of each strip from the ionization chamber as charge. After the signal processing, the position and size of the beam can be fitted and verified by comparing with the physical position of the beam using the shape of the beam spot recorded on medical X-ray film.

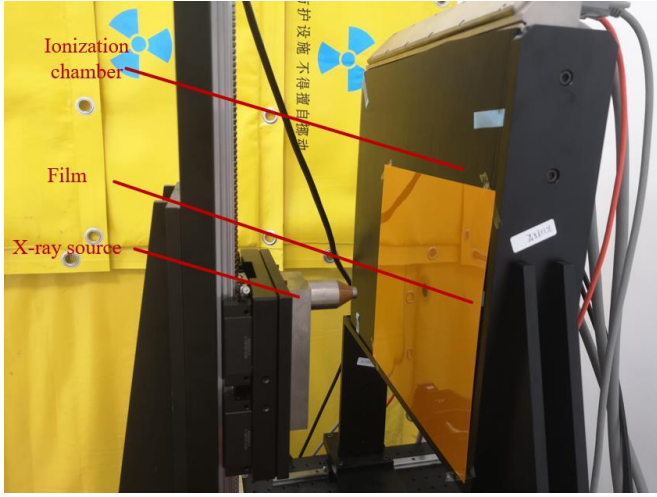


Fig. 4. The relative position of X-ray source, strip ionization chamber and medical X-ray film.

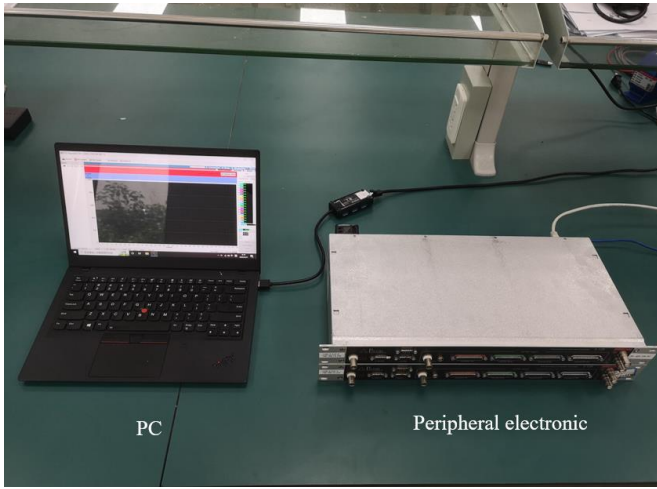


Fig. 5. The peripheral readout electronics of strip ionization chamber (It can integrate the current of each strip from the ionization chamber as charge.).

B. Evaluation of Fitting Curve

In practice, the real position and size of beam are not known exactly, so goodness-of-fit is used to evaluate the fitting effect of Gaussian curve in this study. The goodness-of-fit refers to a statistical test that determines how well sample data fits a distribution from a population with a normal distribution.

Use y to represent the data to be fitted, $\bar{y}$ to represent the average value of the data to be fitted, and $\hat{y}$ to represent the fitting result. The total sum of squares (SST) of observed data is (15).

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 \quad (15)$$

the sum of squares of residuals (SSE) is defined as (16).

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (16)$$

the definition of the coefficient of determination R^2 is (17). R^2 ranges from $-\infty$ to 1, and the closer the R^2 is to 1, the better the fitting effect is [11].

$$R^2 = 1 - \frac{SSE}{SST} \quad (17)$$

C. Data Process

After analyzing the background noise of the experiment setup, the threshold w is set to $7.89e-14C$. Use the modified algorithm to process all strip data in the X and Y directions of the ionization chamber during the beam on, and calculate the mean of the beam center position. Then, the coordinate of the position of the beam is (48.53mm, 85.92mm) relative to the lower right corner of the ionization chamber. In addition, for all the data, the goodness of fit is higher than 0.95, which demonstrates that the algorithm has a good fitting effect. A set of experimental data and fitting result shows in Fig. 6 and Fig. 7, in which the x-coordinate is the strip number.

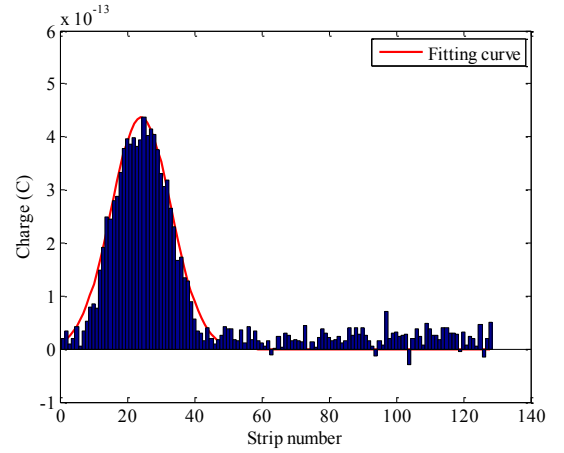


Fig. 6. Fitting result of X-direction experimental data.

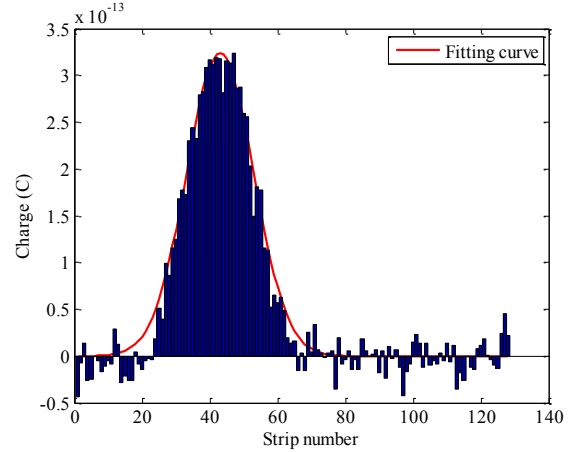


Fig. 7. Fitting results of Y-direction experimental data.

The scan of the medical X-ray film shows that the shape of beam spot is nearly round, as presented in Fig. 8 (a). The medical X-ray film size is 10 inches by 10 inches and the scan resolution is 300dpi, thus the image is made up of 3000 by 3000 pixels. After the film photo is binarized, the contour of the beam is extracted according to the pixel value, as shown in Fig. 8 (b).

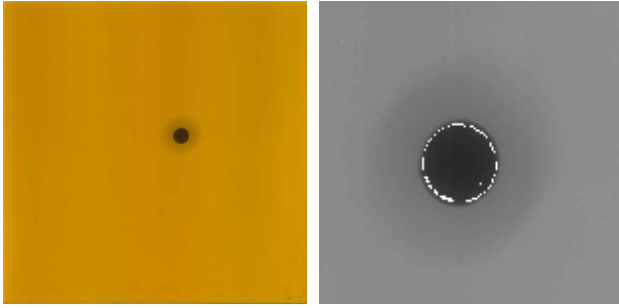


Fig. 8 (a). The shape of the beam presented by medical X-ray film. (b). The contour of the beam.

Use the least square circle fitting algorithm to process the pixel coordinates of the beam outline, and finally obtain the pixel coordinates of the beam spot center is (1347.7, 1759.7). Then, according to the relative position of the film and the strip ionization chamber, the coordinate of the beam is (47.42mm, 84.90mm) relative to the lower right corner of the ionization chamber. By comparing the fitting results of the ionization chamber data with those of the X-ray film, the error is (1.11mm, 1.02mm). Considering that there is some artificial error when fixing the film to the ionization chamber, the fitting result of the algorithm is acceptable.

Considering the possibility of beams shooting on the edge of the ionization chamber, we moved the X-ray source to the edge of the ionization chamber for the second experiment to verify the robustness of the algorithm. At this time, the fitting result obtained by using this method is shown in Figure 9. At this point, the goodness of fit of the fitting curve for all data is higher than 0.8, and for the fitting data is higher than 0.95, which proves that the algorithm is also suitable for this case.

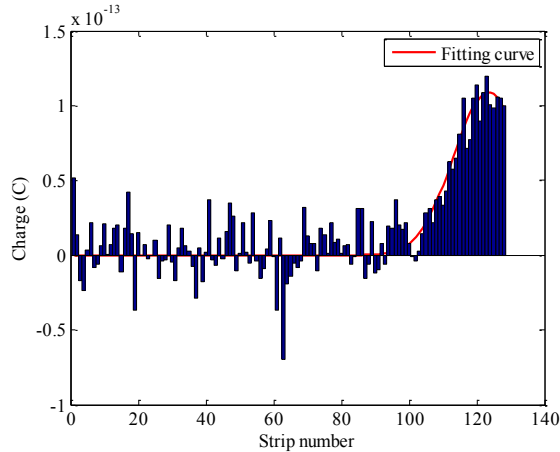


Fig. 9. Fitting results of beam at the edge of ionization chamber

IV. CONCLUSION

Based on the fact that the size of beam spot is small relative to the detection area of ionization chamber in clinical applications, an improved Gaussian fitting method was proposed in this study to obtain the beam center and beam size, which extracts only the strip signals above the noise threshold to reduce the computation and improve the accuracy. The threshold is determined by analyzing the average power of background noise of ionization chamber. By simulation, it was found that for different beam sizes,

when the SNR is higher than 20dB, the accuracy of the beam position is better than 0.2mm, and the relative error of the beam size is better than 2%. And the single execution time of the algorithm is less than 0.4ms under a PC with a CPU frequency of 2GHz. Finally, the accuracy of the proposed algorithm in fitting the beam position is verified by experiments, and it is also demonstrated that this method still works when the beam irradiates at the edge of ionization chamber.

V. ACKNOWLEDGMENT

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